

HANSCOM AFB, MA, USA

WMO: 725059

Lat: 42.470N

Lon: 71.289W

Elev: 41

StdP: 100.84

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 14702

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-13.4	-24.1	0.4	-13.6	-21.9	0.5	-11.4	11.3	-4.1	10.5	-3.2	2.8	290	0.527

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	32.7	22.9	31.1	22.1	29.3	21.2	24.6	30.1	23.8	28.5	22.9	27.0	4.4	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.9	17.7	26.8	22.3	17.1	26.2	21.6	16.3	25.4	74.5	29.9	71.0	28.6	67.6	27.1	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.4	7.5	DB	-20.1	35.3	3.6	1.8	-22.7	36.5	-24.8	37.6	-26.9	38.6	-29.5	39.9	(4)
(5)			WB	-20.6	26.2	3.4	1.0	-23.1	26.8	-25.1	27.4	-27.0	27.9	-29.5	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.8	-3.0	-2.0	2.3	8.7	14.2	19.1	22.5	21.6	17.5	10.9	5.4	0.0
	DBStd	9.98	5.98	5.21	5.18	4.76	4.32	3.91	3.16	3.24	4.07	4.49	4.76	5.14
	HDD10.0	1581	404	337	244	79	8	0	0	0	1	43	152	311
	HDD18.3	3462	662	569	496	291	143	37	4	8	64	234	387	567
	CDD10.0	1523	1	1	7	41	139	274	388	361	224	71	14	2
	CDD18.3	364	0	0	0	3	15	61	134	111	38	3	0	0
	CDH23.3	3437	0	0	3	49	202	585	1278	984	309	27	0	0
	CDH26.7	1157	0	0	0	16	64	185	477	326	86	3	0	0
Wind	WSAvg	3.2	3.4	3.5	3.8	3.8	3.3	3.0	2.7	2.5	2.6	2.9	3.1	3.2
Precipitation	PrecAvg	1169	98	90	105	100	95	89	88	88	96	101	111	111
	PrecMax	1555	336	190	292	237	252	322	224	214	258	337	238	214
	PrecMin	704	18	5	16	21	23	5	13	21	22	15	17	25
	PrecStd	185	60	40	54	47	48	68	42	48	55	68	53	54
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.7	16.7	21.9	29.0	31.4	33.5	35.0	34.1	32.5	26.3	20.7	17.5
		MCWB	13.6	11.7	12.8	17.1	19.7	23.2	24.4	24.1	22.9	18.4	16.0	13.2
	2%	DB	11.7	11.2	17.0	23.5	28.4	31.0	32.8	32.1	29.0	23.3	18.0	13.2
		MCWB	8.8	7.4	10.6	14.1	18.8	22.4	23.1	23.0	21.1	18.0	14.2	10.8
	5%	DB	8.0	8.0	13.2	20.2	25.3	28.7	31.2	30.1	26.9	21.1	16.0	10.5
		MCWB	5.5	4.5	7.9	12.8	17.2	20.7	22.5	21.8	20.0	16.6	12.6	8.0
	10%	DB	5.2	5.6	10.1	17.3	22.6	27.0	29.3	28.2	24.6	18.7	13.6	7.8
		MCWB	2.5	2.3	6.0	11.1	16.0	19.7	21.6	21.0	19.0	14.8	10.5	5.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	13.2	14.7	18.6	21.8	24.9	26.1	25.7	24.1	21.2	17.9	14.8
		MCDB	15.0	16.1	19.3	24.4	29.4	31.1	32.8	31.6	28.6	23.6	19.6	16.4
	2%	WB	9.4	8.1	11.9	16.1	20.2	23.6	24.7	24.5	22.7	19.4	15.5	11.6
		MCDB	11.0	10.1	15.8	22.2	25.7	29.0	30.4	29.5	26.6	22.3	17.3	13.1
	5%	WB	5.5	5.0	8.7	14.2	18.7	22.4	23.8	23.6	21.7	17.4	13.2	8.4
		MCDB	8.1	7.5	12.4	18.8	23.4	26.7	28.7	27.7	24.9	20.4	15.2	10.1
	10%	WB	2.8	2.8	6.4	11.8	17.1	21.1	22.9	22.7	20.6	15.5	10.9	5.8
		MCDB	4.7	5.0	9.6	16.1	21.1	25.0	27.2	26.3	23.5	18.1	13.1	7.7
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.9	10.3	11.5	11.7	11.5	11.6	11.7	11.9	11.3	10.2	9.0
		MCDBR	11.8	12.8	14.8	16.3	16.0	14.0	13.3	13.3	13.5	13.7	12.8	12.0
		MCWBR	9.7	9.6	9.4	8.8	8.1	6.7	5.4	5.5	6.7	8.5	9.4	10.3
	5% WB	MCDBR	11.3	11.7	13.4	14.1	13.7	12.2	11.3	10.9	10.8	11.8	10.9	11.2
		MCWBR	9.9	9.6	9.8	9.1	7.7	6.7	5.4	5.5	6.5	8.6	9.3	10.7
Clear-Sky Solar Irradiance	taub	0.287	0.298	0.324	0.367	0.404	0.446	0.466	0.431	0.372	0.332	0.308	0.291	
	taud	2.485	2.472	2.472	2.385	2.321	2.233	2.185	2.265	2.411	2.537	2.565	2.560	
	Ebn at Noon	865	912	924	901	871	832	810	828	859	861	837	833	
	Edh at Noon	72	86	97	115	126	138	143	128	103	80	67	63	
All-Sky Solar Radiation	RadAvg	1.79	2.65	3.64	4.66	5.14	5.57	5.89	5.30	4.26	2.81	1.93	1.47	
	RadStd	0.11	0.21	0.25	0.50	0.60	0.54	0.36	0.25	0.30	0.25	0.16	0.16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+129	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+92	+42

Nomenclature: See separate page